

Solution of Tutorial-1
Engineering Mechanics: ME-101 (Semester-II, 2024)
Time: 7:55 – 8:50 AM

Q1. The guy cables AB and AC are attached to the top of the transmission tower (**Figure 1**). The tension in the cable AC is 8 kN. Determine the required tension in cable AB such that the net effect of the two cable tensions is a downward force at point A . Determine the magnitude R of the downward force .

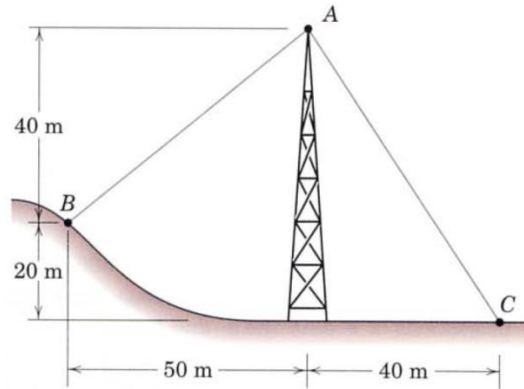
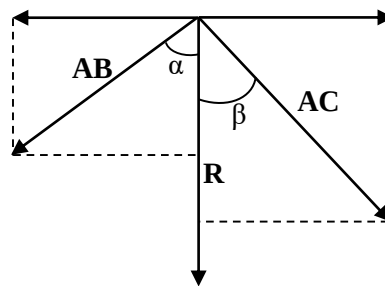
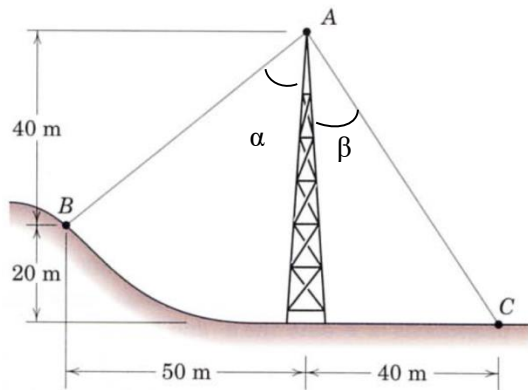


Figure 1

Solution of Q1: **Bold** letters represent vector, while non-bold *italicized* letters represent magnitude



Vector Diagram

2 marks

Let the angles made by the cables AB and AC with vertical be α and β respectively.

From the geometry,

$$\sin \alpha = \frac{50}{\sqrt{50^2 + 40^2}} = 0.781, \cos \alpha = 0.625, \sin \beta = 0.555 \text{ and } \cos \beta = 0.832$$

2 marks

Tensile force in cable AC : $AC = 8$ kN, as given in the question

As the resultant is vertical, the horizontal components of the tensions will add up to zero.

2 marks

Therefore, $AB \sin \alpha = AC \sin \beta$

2 marks

$$\Rightarrow AB = 5.685 \text{ kN}$$

$$\text{Resultant, } R = AB \cos \alpha + AC \cos \beta$$

$$\Rightarrow R = 10.21 \text{ kN.}$$

2 marks

Q2. The t component of the force F is known to be 75 kN and is shown in Figure 2. Determine the n - component and the magnitude of F .

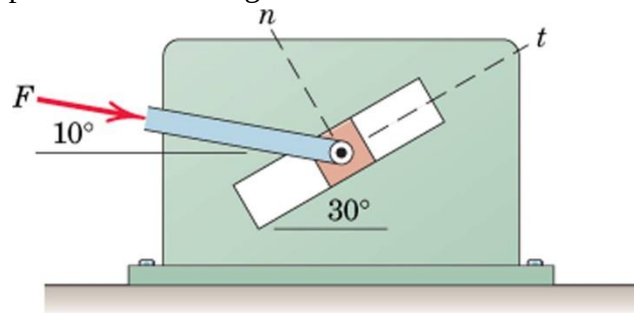
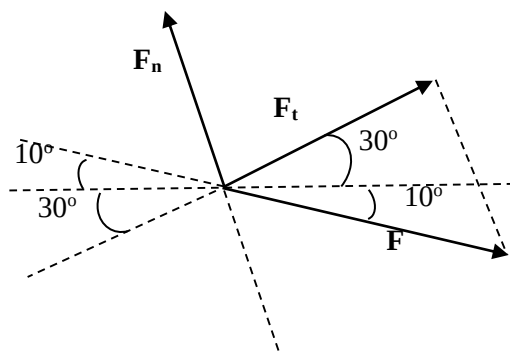


Figure 2

Solution of Q2:



Vector Diagram

2 marks

The angle made by F with the tangential component $F_t = 10^\circ + 30^\circ = 40^\circ$

2 marks

Resolving the components of the forces,

$$F_t = F \cos 40^\circ$$

2 marks

$F_t = 75 \text{ kN}$, as given in the question,

Therefore, $F = 97.91 \text{ kN}$

2 marks

$$F_n = F \cos 130^\circ = -62.9 \text{ kN.}$$

2 marks

Q3a. Two forces F_1 and F_2 act on the hook (Figure 3). If their lines of action are at an angle θ apart and the magnitude of each force is $F_1 = F_2 = F$, determine the magnitude of the resultant force F_R and the angle between F_R and F_1 .

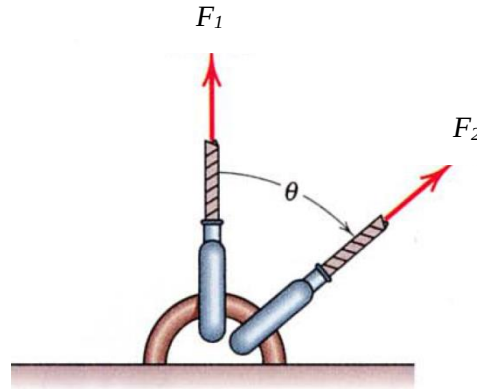
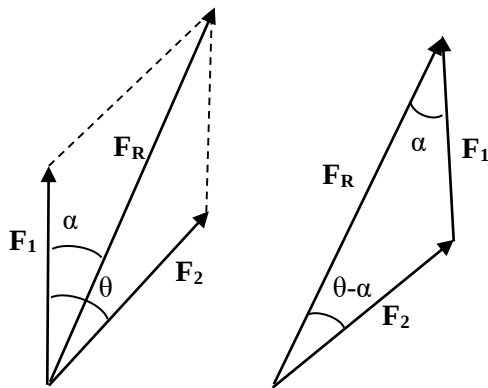


Figure 3a

Solution of Q3:



1 marks

Let α be the angle between F_1 and F_R .

The magnitude of the resultant follows from the law of Cosines:

$$F_R = \sqrt{F_1^2 + F_2^2 - 2F_1F_2 \cos(180^\circ - \theta)}$$

1 marks

As $\cos(180^\circ - \theta) = -\cos\theta$ and $F_1 = F_2 = F$

$$\Rightarrow F_R = \sqrt{F^2 + F^2 + 2F^2 \cos\theta}$$

$$\Rightarrow F_R = F\sqrt{2(1 + \cos\theta)}$$

$$\Rightarrow F_R = 2F\sqrt{\frac{(1 + \cos\theta)}{2}}$$

Since, $\left(\cos\left(\frac{\theta}{2}\right) = \sqrt{\frac{1 + \cos\theta}{2}} \right)$

$$\Rightarrow F_R = 2F \cos\left(\frac{\theta}{2}\right)$$

1 marks

Using Sine Rule:

$$\frac{F_1}{\sin(\theta - \alpha)} = \frac{F_2}{\sin \alpha}$$

1 marks

Since, $F_1 = F_2 = F$

$$\sin(\theta - \alpha) = \sin \alpha$$

$$\Rightarrow \theta - \alpha = \alpha$$

$$\Rightarrow \alpha = \frac{\theta}{2}$$

1 marks

Alternatively,

$$\begin{aligned} \text{As } F_1 = F_2, \quad & \Rightarrow \theta - \alpha = \alpha \\ & \Rightarrow \alpha = \frac{\theta}{2} \end{aligned}$$

Therefore, angle between F_1 and $F_R = \theta/2$.

Question 3b. A 20 N horizontal force is applied perpendicular to the handle of the socket wrench. Determine the magnitude and the coordinate direction angles of the moment created by this force about point O.

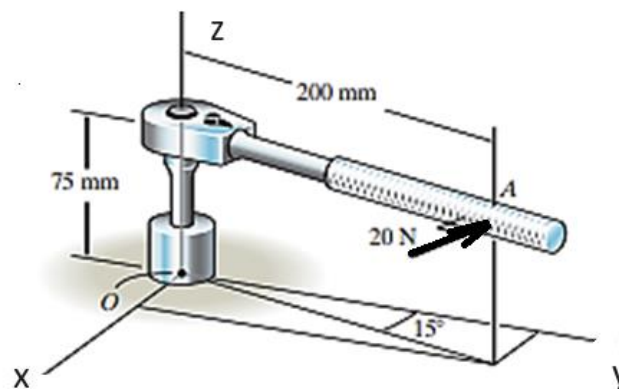


Figure 3b

Solution:

From the given Figure we have $\mathbf{r}_A$ = displacement vector between OA

$\mathbf{F}$ = applied force vector at point A

$\mathbf{M}_o$ = Moment about O

$$\begin{aligned}\mathbf{r}_A &= 0.2 \sin 15^\circ \mathbf{i} + 0.2 \cos 15^\circ \mathbf{j} + 0.075 \mathbf{k} \\ &= 0.05176 \mathbf{i} + 0.1932 \mathbf{j} + 0.075 \mathbf{k}\end{aligned}$$

1 marks

$$\begin{aligned}\mathbf{F} &= -20 \cos 15^\circ \mathbf{i} + 20 \sin 15^\circ \mathbf{j} \\ &= -19.32 \mathbf{i} + 5.176 \mathbf{j}\end{aligned}$$

1 marks

So, for calculating moment about O, we will perform cross-product of position vector and force vector as follows:

$$\begin{aligned}\mathbf{M}_o = \mathbf{r}_A \times \mathbf{F} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0.05176 & 0.1932 & 0.075 \\ -19.32 & 5.176 & 0 \end{vmatrix} \\ &= \{-0.3882 \mathbf{i} - 1.449 \mathbf{j} + 4.00 \mathbf{k}\} \quad \text{N-m}\end{aligned}$$

The magnitude of moment comes out to be-

$$M_o = 4.272 = 4.27 \text{ N-m}$$

1 marks

So, coordinate direction angles of the moment created by this force about point O utilizing direction cosines can be found out by:

$$\alpha = \cos^{-1} \left(\frac{-0.3882}{4.272} \right) = 95.2^\circ$$

$$\beta = \cos^{-1} \left(\frac{-1.449}{4.272} \right) = 110^\circ$$

$$\gamma = \cos^{-1} \left(\frac{4}{4.272} \right) = 20.6^\circ$$

2 marks

Q4. While sliding a desk, three students exert the forces shown in the overhead view in **Figure 4**. Determine the equivalent force—couple system at point A. Then determine the equation of the line of action of the resultant force.

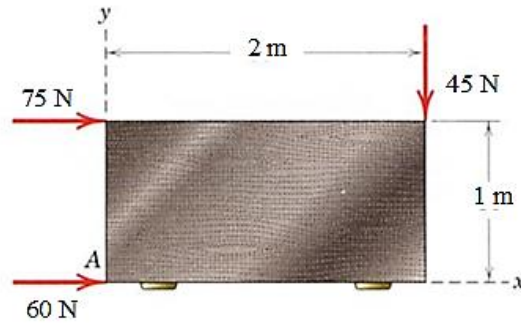


Figure 4

Solution of Q4:

Equivalent force-couple system at point A:

Using the data obtained from figure 2,

✓ **Step 1:** Resultant force along x and y direction are calculated as follows: **[2 Marks]**

$$R_x = (75 + 60) \text{ N} = 135 \text{ N}$$

$$R_y = -45 \text{ N}$$

$$\vec{R} = R_x \hat{i} + R_y \hat{j} = (135\hat{i} - 45\hat{j}) \text{ N}$$

✓ **Step 2:** Resultant moment calculation: **[3 Marks]**

Let us consider counter-clockwise moment as positive and clockwise moment as negative.

From figure 2, taking moment about A we get:

$$\begin{aligned} \vec{M}_A &= (-75 \times 1 - 45 \times 2) \hat{k} \text{ N.m} \\ &= -165 \hat{k} \text{ N.m} \end{aligned}$$

Thus, equivalent force and couple system is

$$\vec{R} = (135\hat{i} - 45\hat{j}) \text{ N}$$

$$\vec{M}_A = -165 \hat{k} \text{ N.m}$$

✓ **Step 3: Condition for line of action of resultant**

Let, $\vec{r} = x\hat{i} + y\hat{j}$ is a position vector running from point A to any point on the line of action of

R. Substituting the vector expressions for $\vec{r}$, $\vec{R}$, and $\vec{M}_A$, and carrying out the cross-product

result in,

$$\vec{r} \times \vec{R} = \vec{M}_A$$

$$\Rightarrow (x\hat{i} + y\hat{j}) \times (135\hat{i} - 45\hat{j}) = -165\hat{k}$$

$$\Rightarrow (-45x - 135y)\hat{k} = -165\hat{k}$$

$$\Rightarrow -45x - 135y = -165$$

$$\Rightarrow -135y = 45x - 165$$

$$\Rightarrow y = (-0.33x + 1.22) \text{ m}$$

[5 Marks]

Q5. The structure as shown in **Figure 5** is constructed of circular rod which has a mass of 7 kg per meter of length. Determine the moment M_O about O caused by the weight of the structure. Find the magnitude of M_O . [5+5=10 marks]

Solution of Q5:

Step 1: By using Figure 5.1, the load W_1 in section **AB** can be calculated as below:

$$W_1 = 2.2 \times 7 \times 9.81 = 151 \text{ N} \quad (1 \text{ mark})$$

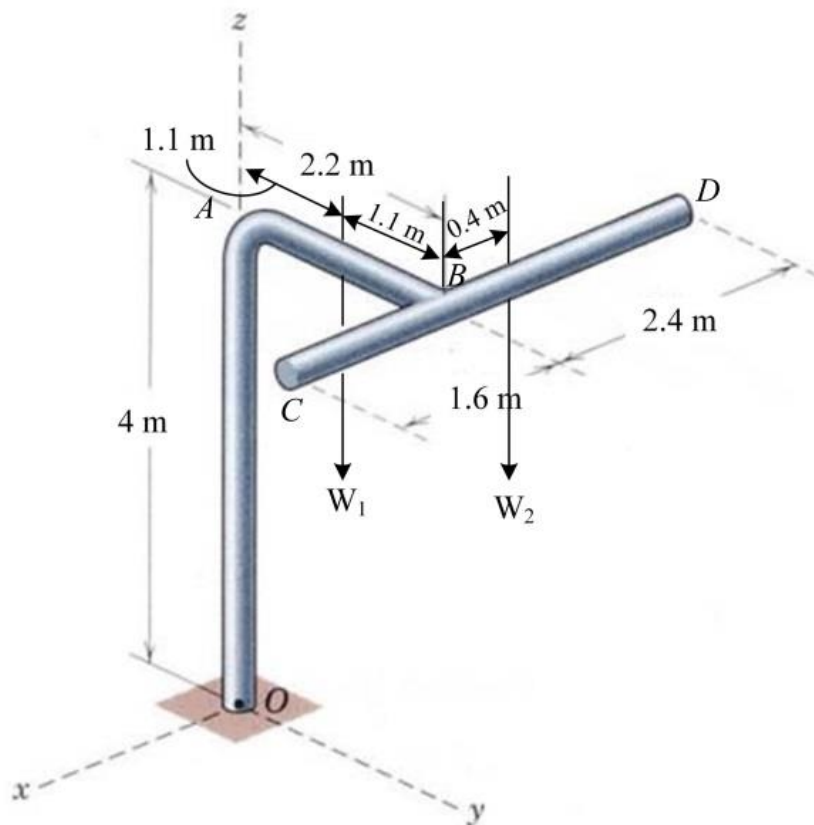


Figure 5.1

Step 2: By using Figure 5.1, the load W_2 in section **CD** can be calculated as below:

$$W_2 = 4 \times 7 \times 9.81 = 274.68 \text{ N} \quad (1 \text{ mark})$$

Step 3: Load W_1 is acting at centre point of **AB** and W_2 is acting at centre point of **CD**

Taking moment about x-axis

$$M_{O_x} = -151(1.1) - 274.68(2.2) = -770.4 \text{ N.m} \quad (2 \text{ marks})$$

Step 4: Taking moment about y-axis

$$M_{O_y} = -274.68(0.4) = -109.87 \text{ N.m} \quad (2 \text{ marks})$$

Step 5: Taking moment about z-axis

$$M_{O_z} = 0 \quad (1 \text{ mark})$$

Step 6: Resultant moment vector on point O

$$M_O = -770.4 \hat{i} - 109.87 \hat{j} \text{ N.m} \quad (1 \text{ mark})$$

Step 7: Resultant moment on point O

$$M_O = \sqrt{(-770.4)^2 + (-109.87)^2} = 778.2 \text{ N.m} \quad (2 \text{ marks})$$

Q6. A uniform wheel of 800 mm diameter of weight 8 kN rests against a rigid rectangular block of 200 mm height as shown in **Figure 6**.

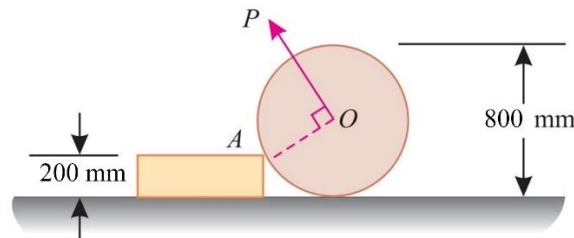


Figure 6

Find the least pull, through the centre of the wheel, required just to turn the wheel over the corner **A** of the block. Also find the reaction on the block at **A**. Consider all the surfaces to be smooth i.e., frictionless. **[5+5= 10 marks]**

Solution of Q6:

Given: Diameter of wheel = 800 mm; Weight of wheel = 8 kN and height of the block = 200 mm.

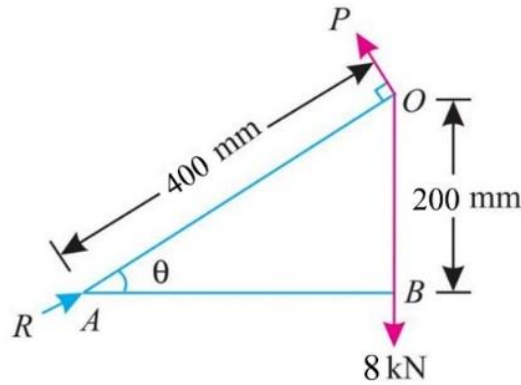


Figure 6.1

Least pull required just to turn the wheel over the corner.

Step 1: Let us consider, P = Least pull required just to turn the wheel in kN.

A little consideration will show that for the least pull, it must be applied normal to AO. The system of forces is shown in figure 6.1. From the geometry of the figure, we find that

$$\sin \theta = \frac{200}{400} = 0.5 \text{ or } \theta = 30^\circ \quad (1 \text{ mark})$$

And $AB = \sqrt{(400)^2 - (200)^2} = 346.4 \text{ mm} \quad (1 \text{ mark})$

Step 2: Calculation of least pull required just to turn the wheel over the corner.

Now taking moments about A and equating the same,

$$P \times 400 = 8 \times 346.4 = 2771.2$$

$$\therefore P = \frac{2771.2}{400} = 6.93 \text{ kN} \quad (4 \text{ marks})$$

Step 3: Calculation of Reaction on the block:

Let us consider, R = Reaction on the block in kN.

Resolving the forces horizontally and equating the same,

$$R \cos 30^\circ = P \sin 30^\circ$$

$$R = \frac{P \sin 30^\circ}{\cos 30^\circ} = \frac{6.93 \times 0.5}{0.866} = 4 \text{ kN} \quad (4 \text{ marks})$$